Sr.No. 6565

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26 | Session: Odd MSE October 2025 | Date: 08-10-2025
Sem: 1 | Slot: 10:00am - 11:00am | C. Code: CS413
Exam Type: Written | Block: 202 | Course Section: 1 | Seat No: 2022600007
1_61_465_986_1_1505_48215

Name of the candidate. Bhoi Kunal Shantaram

Candidate's UID number: 2 0 2 2 6 0 0 0 0 7

Examination class room number: 2 0 2

Invigilator's signature with date: 08/10/2025

(To be filled in by the candidate)

EXAMINATION: BY BTECH (MSE)
(For example FY MCA/FY BTECH)

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE (AIML)

DAY AND DATE: Wednesday 08/10/2025

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

RE/2025/03

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

Q2(a) → For standard 2D convolution :

if input is $H \times H$ and filter is $F \times F$
output will be $(H-F+1) \times (H-F+1) \times K$
because we have K filters and
$(H-F+1) \times (H-F+1)$ will be size of one filter.
b/coz of

And with depth here considering C we
require filter with size $F \times F \times C$

Hence output will be of $(H-F+1) \times (H-F+1) \times K$

No. of parameters = (iput_n x kernel_n)
* (kernel_n * kernel_n) +
kernel_n

depthwise are critical as they calculate
otput for current layer using channels
rather than black and white images
like 3 channels for rgb.

So CNN learns features by using channels
data too hence better in many cases.

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Q1(a) → $f(w) = \frac{1}{2} \|Xw - y\|^2$.

in gredient update rule weight updates according to

$w_i \leftarrow w_i + \Delta w_i$

where $\Delta w_i = -\eta \frac{\partial f(w)}{\partial w}$

∴ In our case $f(w) = \frac{1}{2} \|Xw - y\|^2$

$f'(w) = \frac{1}{2} \frac{\partial}{\partial w} (Xw - y)^2$

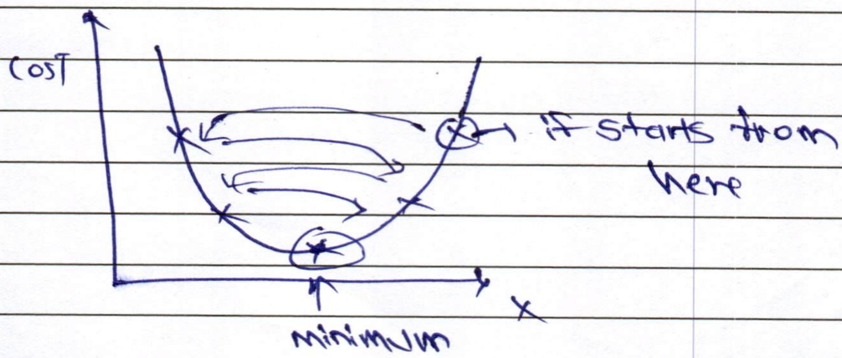
$= \frac{1}{\cancel{2}} \times \cancel{2} (Xw - y) \frac{\partial}{\partial w} (Xw - y)$

$= (Xw - y) * X$

∴ gredient update rule will be

$x_{k+1} = x_k - \eta (Xw - y) x$

new weight   learning rate



if learning rate will be $\geq \frac{2}{\lambda_{max}}$

we will get next x value at opposite side of minimum then again same thing happens as shown in graph.

Her $\eta < \frac{2}{\lambda_{max}}$ wher λ_{max} is largest eigenvalue of $X^T X$

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	Q2.b →	linear autoender with input x

encoder ⇒ $z = f(x)$ ← ~~output means~~ compressed input in latent space

Decoder ⇒ $\hat{x} = g(z)$ ← output (original input from compressed input)

$L = \|x - \hat{x}\|^2$ → should be ~~as~~ minimum but compresed input must contain important features.
↑ Loss

$z = f(x)$

$\therefore z = Wx$ → W is weight matrix for encoder

$x \in \mathbb{R}^n$

~~$W = \begin{pmatrix} I_n \\ 0_{(k-n)\times n} \end{pmatrix}$~~

$W = \begin{pmatrix} I_n \\ 0_{(k-n)\times n} \end{pmatrix}$ → This will give same input x in latent space

$z = \begin{pmatrix} I_n \\ 0 \end{pmatrix} (x \ 0) = x$

Decoder :

$\hat{x} = g(z)$

$= U\hat{x}$ → U is weight matrix for decoder

$U = (I_n \ 0_{(k-n)\times n})$

$\therefore \hat{x} = (I_n \ 0_{k-n\times n}) \begin{pmatrix} x \\ 0 \end{pmatrix} = x$ — ①

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$\therefore \text{Loss} = \|x - \hat{x}\|^2$

$= \|x - x\|^2$ .... from eqn ①

$= 0$

Hence proved ~~it~~ 0 loss means it learned ~~identity function~~. same ~~th~~ input as it is and did not understand underlying patterns.

Hence regularization is needed so it will understand underlying patterns.

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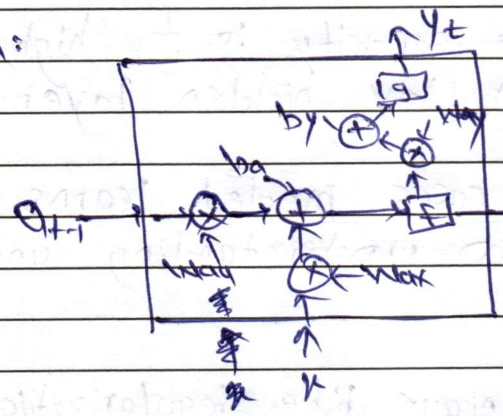
Q3a → RNN:



$$a_t = \text{function}(W_{aa} \times a_{t-1} + W_{ax} \times x + b_a)$$

$$y_t = g(W_{ay} \times a_t + b_y)$$

Forget gate $f_t = \sigma(W_f [h_{t-1}, x_t] + b_f)$
Here if value is $0 \to$ means don't keep
$1 \to$ keep everything
Hence forget gate will remove things as it see 0 in matrix.

And in LSTM we have cell state

$$C_t = (f_t * C_{t-1}) + (i_t * \tilde{C}_t)$$

this cell state will carry old information (input) and prevent it from vanishing as eq^n. contain forget gate.

input gate acts for if want to add input in current context or not.

$$i_t = \sigma(W_i [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_c [h_{t-1}, x_t) + b_{\tilde{c}})$$

$$O_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$h_t = C_t * \tanh(O_t) \to$ output at hidden state.

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	q.1.b→	if models capacity is too high like more neurons at ~~lay~~ hidden layer.

In such cases model learns training data rather than understanding underlying patterns.

Here techniques like regularization will help as it active some neurons so model can learn underyling patterns.

As model ~~l~~learns traning data it will perform best on traning input when provided. Hence high accuracy on traning data inputs.

But low accuracy on unseen data as it learned traned data and never seen unseen data.

Hence overfitting occures when model capacity is too high.

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Q3b → GRU:

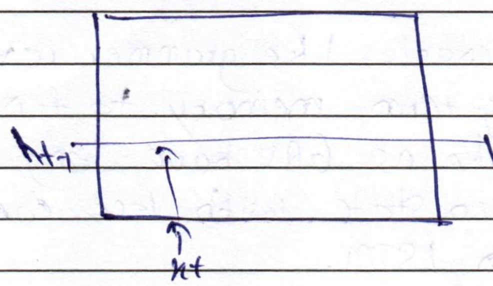
it has 2 gates

① update gate : $u_t = \sigma(W_u[h_{t-1}, x_t] + b_u)$

→ responsible for what to update means what to keep

② reset gate: $r_t = \sigma(W_r[h_{t-1}, x_t] + b_r)$

→ responsible for what to forget and don't keep



h_t → hidden state it keeps context of old inputs which are important.

$h_t = (u_t * \tilde{c}_t) + ((1-u_t) * h_{t-1})$

candidate term

$\tilde{c}_t = \tanh(W_c[r_t * h_{t-1}, x_t] + b_c)$

update rule.

it is combination of previous hidden state h_{t-1} and candidate state $\tilde{c}_t$.

LSTM have 4 terms:
input gate, output gate, forget gate, candidate state

while GRU have 3 terms:
reset gate, update gate, candidate state.

Hence number of parameters are less in GRU as it have 3 terms while LSTM have 4.

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Hence computations in GRU are less that of LSTM.

Because of less computational complexity at each timestamp GRU can be better than LSTM in below scenarios:

1. Named-entity recognition
2. Baby-name generation
3. grammer correction
4. Language generation.

More better in scenario like grammer correction as it need long-term memory to find gramatical mistake as GRU have long-term memory in hidden state with less computational complexity than LSTM.

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